DESIGN AND PERFORMANCE ANALYSIS OF A DC–DC BUCK CONVERTER USING LTspice

Independent Electrical Engineering Simulation Project

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Software: LTspice

Project Type: Independent simulation-based project

ABSTRACT

This project investigates the design and simulated performance of a DC–DC buck converter using LTspice. A buck converter is a switching power-electronic circuit that reduces a DC input voltage to a lower DC output voltage. Such converters are commonly used in battery-powered devices, embedded systems, voltage regulators, and other electronic equipment.

The investigation examines the effects of duty cycle, switching frequency, inductance, and capacitance on converter behavior. Transient simulations are used to observe output voltage and current waveforms and to estimate current and voltage ripple. Input and output power are also investigated to understand the concept of conversion efficiency.

The converter was simulated using a MOSFET, diode, inductor, capacitor, and resistive load. A reported input voltage of 12 V and an output voltage of approximately 4.89 V were obtained under the baseline operating conditions. Additional experiments explored changes in duty cycle, switching period, inductance, and capacitance.

The project demonstrates how circuit simulation can be used to investigate switching-converter behavior, compare theoretical predictions with simulated results, and identify the effects of component selection on power-conversion performance. The findings are limited to the accuracy of the circuit model and the measurements obtained in simulation.

1. INTRODUCTION

Electrical and electronic systems often require a stable DC voltage that differs from the voltage supplied by a battery, power adapter, or another DC source. DC–DC converters provide a method of converting one DC voltage level into another.

A buck converter is a step-down switching converter. It uses a controlled switching device, an energy-storage inductor, a diode or equivalent current path, and a capacitor to produce a lower DC output voltage.

Unlike a simple resistive voltage divider, a switching converter transfers energy through controlled switching and energy-storage components. Its performance depends on operating conditions and component parameters.

In this project, LTspice is used to construct and simulate a buck-converter circuit. Different operating parameters are varied to investigate their effects on output voltage, current ripple, and estimated conversion efficiency.

The project provides practical experience in circuit simulation, measurement, data analysis, and the application of electrical engineering theory.

2. PROJECT OBJECTIVES

The objectives of this project are:

1. To construct and simulate a DC–DC buck converter using LTspice.
2. To investigate the relationship between duty cycle and average output voltage.
3. To examine the effect of switching frequency on converter waveforms.
4. To investigate the relationship between inductance and inductor current ripple.
5. To examine the effect of output capacitance on output-voltage ripple.
6. To estimate input power, output power, and conversion efficiency.
7. To compare simulated behavior with theoretical predictions.
8. To develop practical skills in circuit design, simulation, troubleshooting, and technical reporting.

3. THEORETICAL BACKGROUND

3.1 Principle of operation

A buck converter controls the transfer of energy from a DC source to a load by rapidly switching a semiconductor device on and off.

During the switch-on interval, energy is transferred from the input through the inductor to the output. During the switch-off interval, the inductor continues to supply current through the circuit's freewheeling path.

The output capacitor helps reduce voltage fluctuations, while the inductor limits rapid changes in current.

3.2 Duty cycle and output voltage

Duty cycle is the fraction of each switching period during which the switching device is on.

It is defined as:

$D = Ton / Ts$

where:

- D is the duty cycle.
- Ton is the switch-on time.
- Ts is the total switching period.

For an ideal buck converter operating in steady state under continuous-conduction conditions, the average output voltage is approximately:

$$V_{out} = D \times V_{in}$$

For an input voltage of 12 V and a duty cycle of 0.40:

$$V_{out} = 0.40 \times 12 = 4.8 \text{ V}$$

This theoretical value provides a reference for comparison with the simulated output voltage.

3.3 Switching frequency

Switching frequency is the number of switching cycles per second:

$$f_s = 1 / T_s$$

For a switching period of 40 microseconds:

$$f_s = 1 / (40 \times 10^{-6}) = 25 \text{ kHz}$$

Increasing switching frequency can reduce current ripple under otherwise comparable ideal operating conditions. However, switching losses may also increase in a practical converter.

3.4 Inductor current ripple

The inductor stores energy in its magnetic field and limits changes in current. Current ripple is the difference between the maximum and minimum inductor current over a switching cycle:

$$\Delta I_L = I_{L(max)} - I_{L(min)}$$

A larger inductance generally reduces current ripple when other operating conditions remain constant.

3.5 Output-voltage ripple and capacitance

The output capacitor stores electrical energy and helps maintain the output voltage between switching events.

Output-voltage ripple is defined as:

$\Delta V_{out} = V_{out(max)} - V_{out(min)}$

For otherwise comparable operating conditions, increasing capacitance generally reduces the capacitive component of output-voltage ripple. Equivalent series resistance, circuit topology, and transient behavior can also influence the measured ripple.

3.6 Conversion efficiency

Conversion efficiency expresses the fraction of input power delivered to the output:

$\eta = (P_{out} / P_{in}) \times 100\%$

where Pin is input power and Pout is output power.

The power values should ideally be measured as average instantaneous power over the same steady-state interval:

$P_{in} = \text{average}[V_{in} \times I_{in}]$

$P_{out} = \text{average}[V_{out} \times I_{out}]$

The current direction and sign convention must be handled consistently. A realistic efficiency estimate also depends on whether the simulation includes semiconductor conduction losses, switching losses, inductor resistance, and other non-ideal effects.

4. CIRCUIT DESIGN AND SIMULATION METHOD

The converter was constructed in LTspice using a DC input source, MOSFET switching device, diode, inductor, output capacitor, and resistive load.

The circuit was analyzed using transient simulation to observe how voltage and current changed with time.

Baseline operating conditions

Parameter	Baseline setting
Input voltage	12 V
Duty cycle	40%
Switching period	40 μs
Switching frequency	25 kHz
Transient duration	5 ms
Switching device	MOSFET
Freewheeling path	Diode
Energy-storage components	Inductor and capacitor
Load	Resistive load

The inductor, capacitor, load resistance, and semiconductor model values should be documented using the actual values in the saved LTspice schematic.

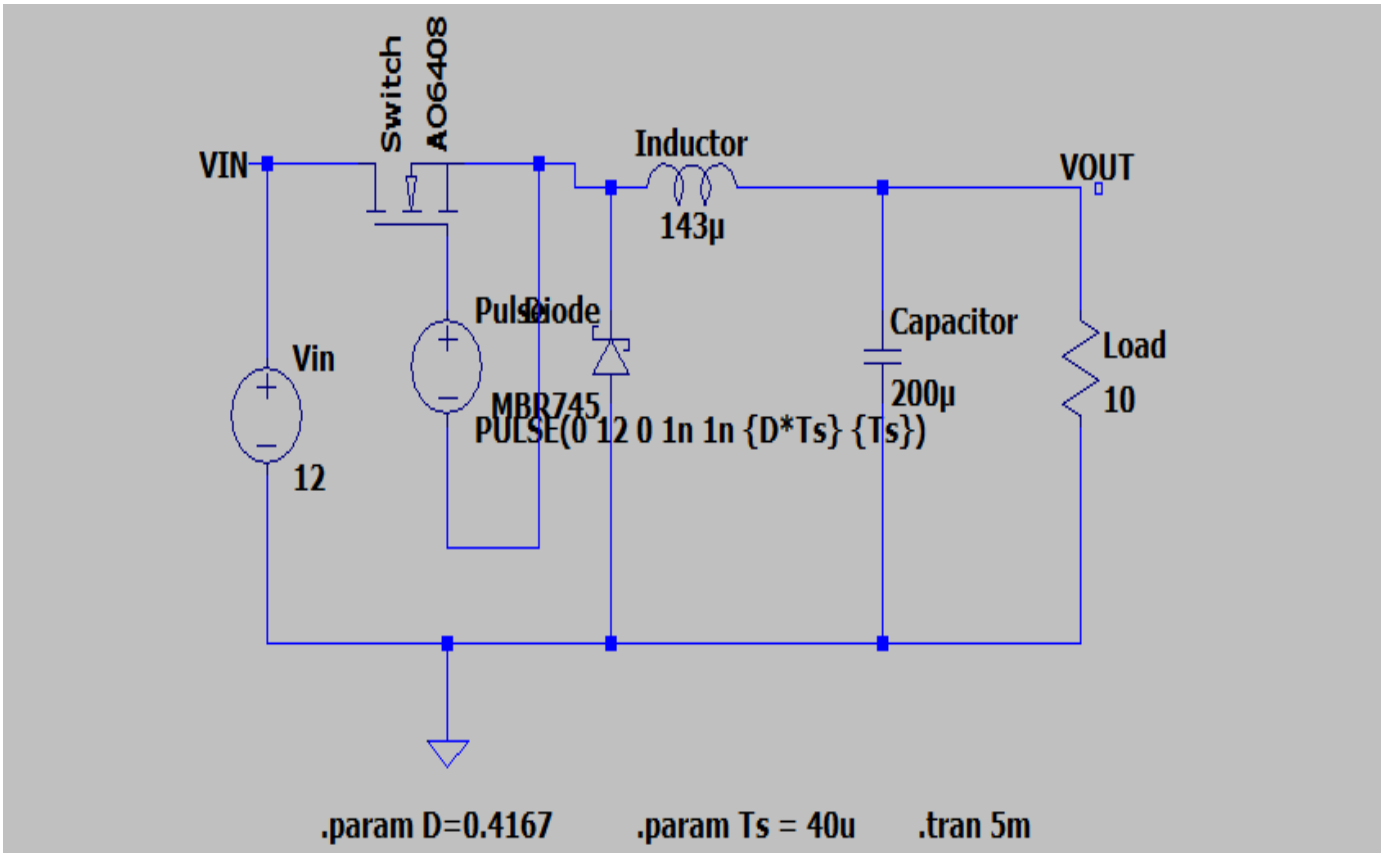
Experimental procedure

The circuit was first simulated under the baseline conditions. One parameter was then varied at a time while the other parameters were kept constant.

The duty cycle, switching period, inductance, and capacitance were investigated separately. Waveforms were inspected after the initial transient, with measurements intended to use the interval from 4.5 ms to 5 ms where the circuit had reached steady state.

The measurements were recorded in tables and compared with theoretical expectations. The investigation was simulation-based.

The given figure below shows the Circuit used in the Ltspice Based Simulation:



5. RESULTS AND OBSERVATIONS

5.1 Duty-cycle investigation

The duty cycle was varied between 30% and 60%.

Duty cycle	Ideal output voltage for 12 V input	Recorded observation
30%	3.6 V	Approx
40%	4.8 V	Approximately 4.9 V reported
50%	6.0 V	Final numerical measurement
60%	7.2 V	Approximately 7.1 V reported near 4.95 ms

The results reported for 40% duty cycle is close to the ideal buck-converter relationship. This supports the theoretical expectation that duty cycle strongly influences the average output voltage.

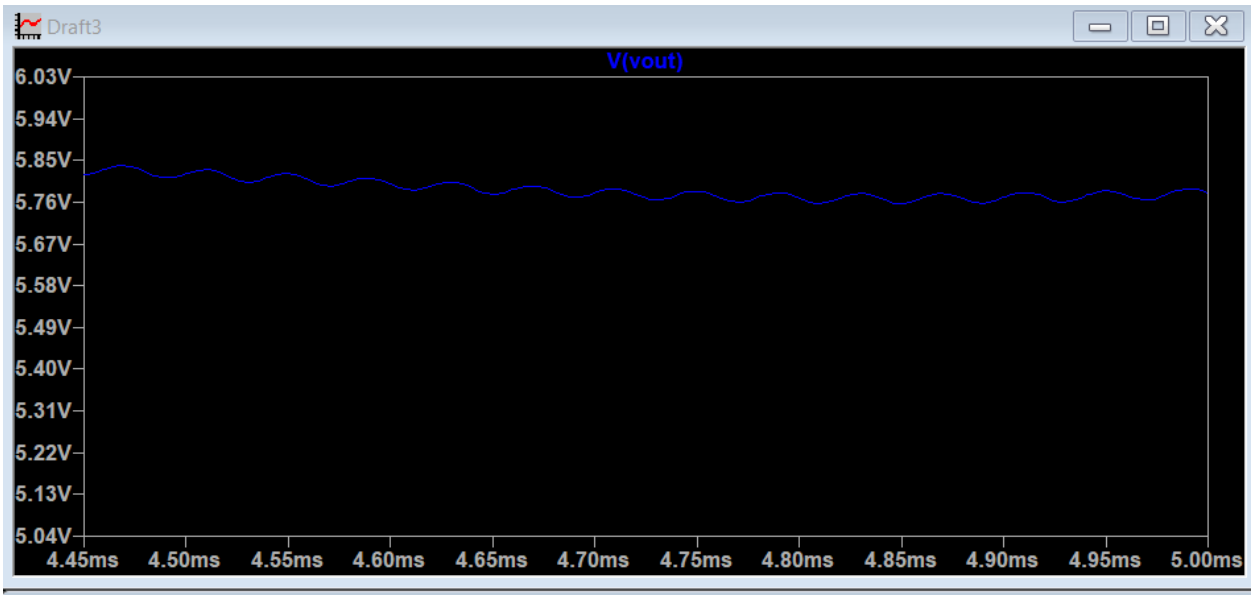


Figure: Voltage out for 50% Duty cycle

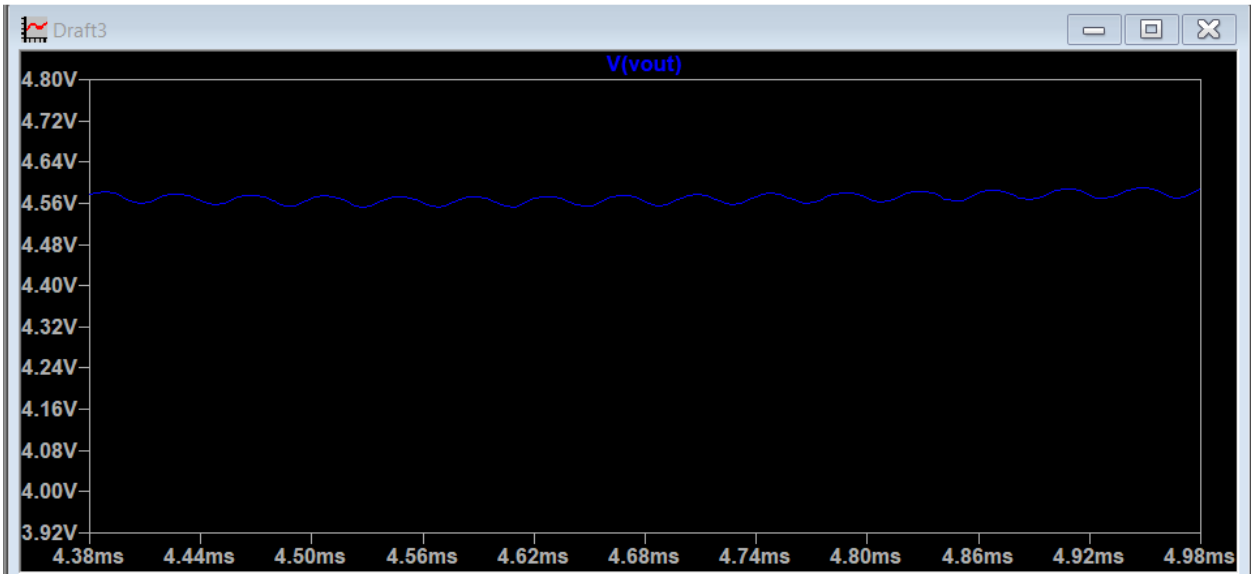


Figure: Voltage out for 40% Duty Cycle

5.2 Switching-frequency investigation

The switching period was changed while the other circuit settings were intended to remain constant.

Switching period	Switching frequency	Result
40 µs	25 kHz	Simulated
20 µs	50 kHz	Simulated
10 µs	100 kHz	Simulated

The output voltage was reported to change relatively little across these tests. This is consistent with the fact that the ideal average output voltage is primarily determined by duty cycle and input voltage. Frequency-related effects are more clearly investigated by comparing ripple, switching losses, and other waveform characteristics.

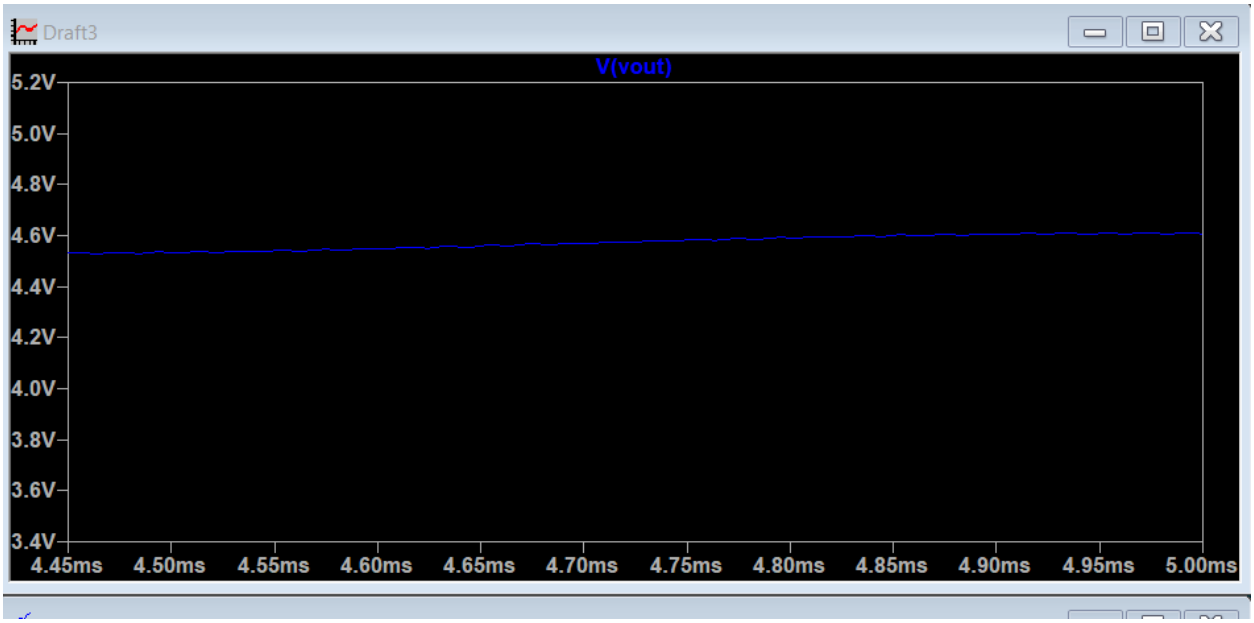


Figure: Voltage out for all 3 changes in frequency

5.3 Inductance investigation

The inductor current ripple was calculated using the difference between the peak and valley current.

	Inductance	Peak current	Valley current	Calculated ripple
1	68 µH	0.317 A	0.310 A	0.007 A
2	143 µH	0.313 A	0.309 A	0.004 A
3	370 µH	0.319 A	0.316 A	0.003 A

The recorded ripple decreases across these three runs. This is consistent with the expected effect of increasing inductance, provided the other operating conditions are comparable. The inductance values for runs 2 and 3 must be confirmed before drawing a final quantitative conclusion.

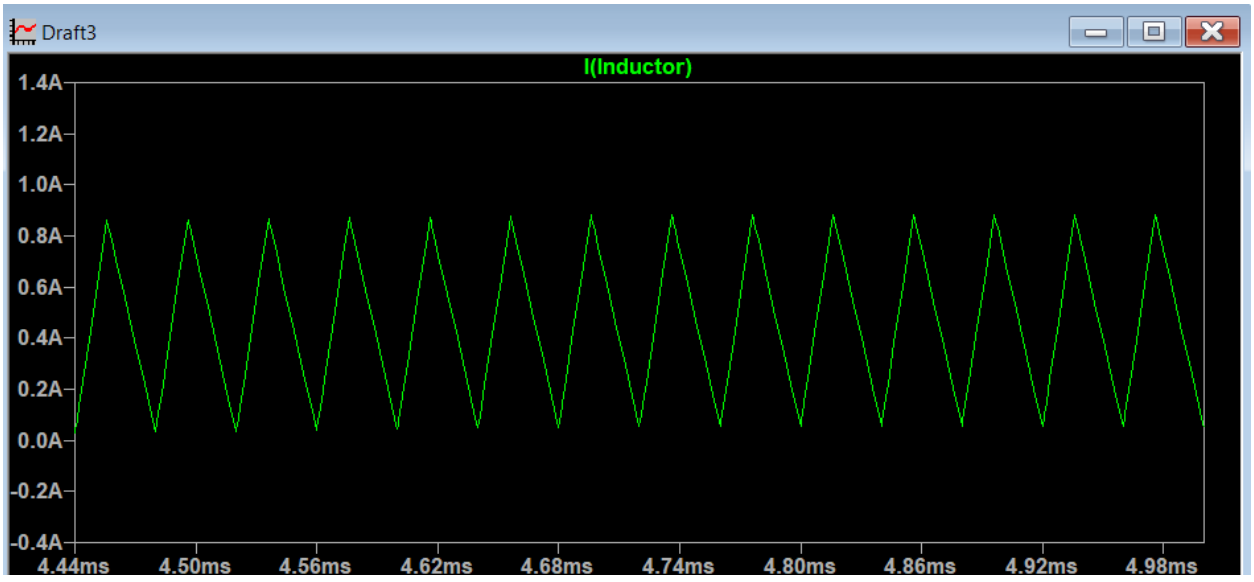


Figure: Current for inductor with inductance 143 µH

5.4 Capacitance investigation

The following readings were recorded during the capacitance tests.These are from 4.5 to 5ms of timings.

Capacitor setting	Highest reported reading	Lowest reported reading	Voltage Ripple
100 µF	3.22V	3.19V	26mV
200 µF	3.21V	3.20V	10mV
450 µF	3.14V	3.11V	33mV

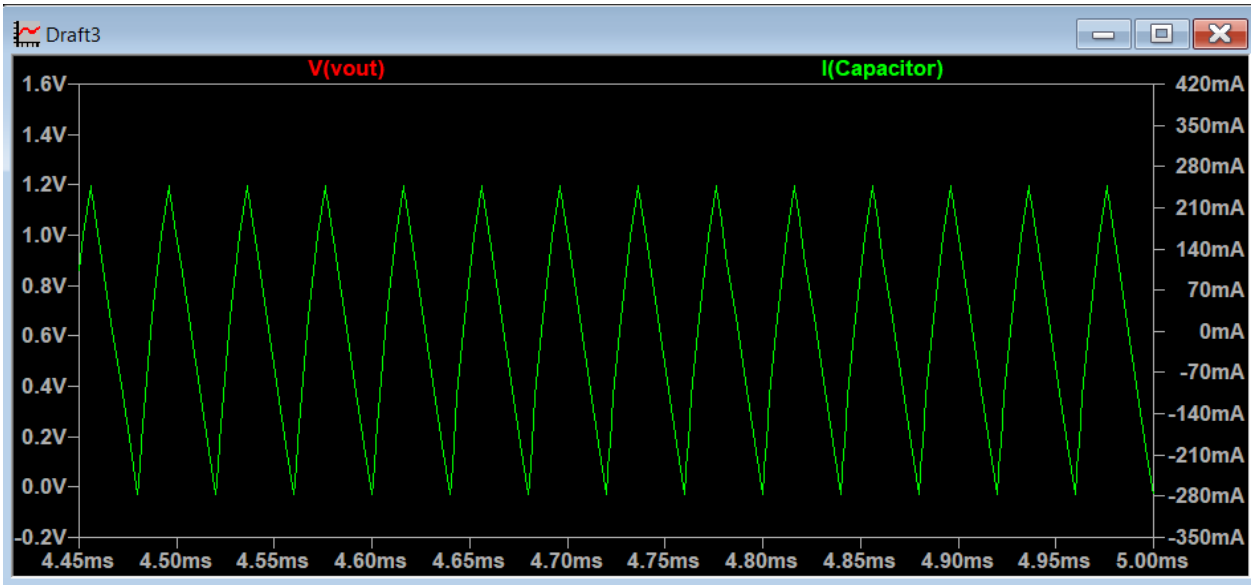


Figure: Voltage out for 200µ Capacitance

5.5 Output Power and Conversion Efficiency

Table 5.5: Efficiency Calculation

Parameter	Value
Input voltage	12 V
Input current	0.135A
Output voltage	4.89 V
Output current	0.318A
Input power	1.620 W
Output power	1.558 W
Conversion efficiency	**96.16%**

Calculations

Input power = Input voltage × Input current

= 12 × 0.135= 1.620 W

Output power = Output voltage × Output current

= 4.89 × 0.318 = 1.558 W

Efficiency = (Output power ÷ Input power) × 100%= (1.620 ÷ 1.558) × 100%
Efficiency= 96.16**%**

6. CONCLUSION AND EVERYDAY HOUSEHOLD APPLICATIONS

The project investigated the design and performance of a DC–DC buck converter using LTspice simulation. By examining duty cycle, switching frequency, inductance, capacitance, and power-conversion efficiency, the study explored how the main operating parameters influence the performance of a switching power converter.

The simulation produced an output voltage of approximately 4.89 V from a 12 V DC input under the reported baseline conditions. This is close to the theoretical output of 4.8 V at a 40% duty cycle for an ideal buck converter. The comparison demonstrates how simulation can be used to examine electrical engineering theory and understand the behavior of practical circuit designs.

Everyday Applications and Typical Voltage Requirements

DC–DC buck converters are widely relevant to household electronics because different devices require different DC voltage levels. A converter can reduce a higher DC voltage to a lower, more suitable level when its input range, output-current rating, and protection features are designed for the application.

Household application	Typical DC voltage or supply requirement	Possible role of a buck converter

USB-powered electronics	Common USB output: 5 V DC	Reduce a higher DC voltage to a regulated 5 V supply when appropriately designed
Wi-Fi routers	Often 5 V or 9 V DC, depending on the model	Provide a suitable lower DC voltage from a compatible higher-voltage source
LED lighting systems	Depends on the LED arrangement and driver design	Reduce voltage or regulate current using a suitable converter or LED driver
Battery-powered devices	Depends on the battery and device	Adapt battery voltage to the voltage required by internal electronics
Solar-powered household systems	Depends on the panel, battery, and connected equipment	Regulate DC voltage for compatible loads or charging-system stages
Electronic control circuits	Commonly 3.3 V or 5 V DC	Supply lower voltage rails for sensors, microcontrollers, and control boards

These values are typical examples rather than universal specifications. Actual voltage and current requirements must be checked on the device label or manufacturer documentation. A buck converter cannot increase voltage, and household AC mains must not be connected directly to a low-voltage DC converter; a suitable, isolated AC–DC power supply is required where conversion from mains electricity is involved.

Practical Significance of the Investigation

The results highlight the importance of selecting appropriate components for reliable voltage conversion. Duty cycle influences the average output voltage, inductance affects current ripple, and output capacitance influences voltage ripple. Switching frequency also affects ripple and can influence switching losses in a practical design.

The initial efficiency calculation produced approximately 95.44% using a 12 V input, 0.175 A input current, 4.89 V output, and an assumed output current of 0.410 A. This is an illustrative calculation, not a verified simulation result, because the output current must be confirmed against the actual circuit and load resistance. Reliable efficiency evaluation requires measuring input and output power over the same steady-state interval in LTspice.

Research into typical household DC power requirements also demonstrates why efficient voltage conversion matters. Many electronic devices operate internally at low DC voltages even when their original power source supplies a different voltage. A well-designed converter can help provide the required voltage while reducing unnecessary power loss, although actual efficiency depends on component selection, load conditions, thermal behavior, and circuit design.

Future Improvements

Further investigation could include verifying the efficiency through average-power measurements, testing different MOSFET and diode models, comparing component losses, and selecting component values to reduce current and voltage ripple. A future extension could build a low-voltage physical prototype and compare its measured performance with the LTspice results, using suitable current limiting and electrical safety precautions.

Overall, this project developed an understanding of switching power conversion and demonstrated how circuit simulation can support engineering analysis before hardware is built. It also connected theoretical concepts to everyday electronic systems, where accurate voltage regulation, low ripple, suitable current capacity, and efficient power delivery are important design considerations.